RÉSUMÉ

Mithilesh Gautam

Backend Software Engineer · Bengaluru, India

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Backend software engineer with **4+ years** building and optimizing production SaaS applications with **Python, Django, PostgreSQL and REST APIs**. I work on backend architecture, database performance, asynchronous processing, distributed systems, AI/LLM integrations and legacy modernization, and I deliver features end to end across backend and frontend.

01 Experience

Backend Software Engineer · Gainfront

March 2024 – Present · Bengaluru, India

- Engineered **Python/Django backend services** in a multi-tenant microservice architecture: PostgreSQL models, REST APIs, ORM queries and business workflows across interconnected services.
- Built a **PostgreSQL-native distributed task executor** for background processing, with database-backed queuing, task isolation, fan-out/join workflows and idempotent execution, without Celery or Redis.
- Optimized PostgreSQL analytics with precomputed rollups, 56+ indexes, partial and composite indexing, JSONB and full-text search, cutting measured query time from **239 s to 0.9 s**.
- Engineered an **AI/RAG platform with multi-model LLM routing**: provider abstractions, embeddings, pgvector similarity search, tool orchestration, structured outputs and multi-step agent workflows.
- Modernized legacy Django template modules into **Vue.js + REST**, replacing server-rendered views with decoupled components and API-driven workflows.
- Owned features end to end with clients: turned requirements into technical designs, pushed back on infeasible approaches with alternatives, shipped tested implementations and resolved production issues through Sentry.

Backend Software Engineer · MarsDevs

March 2022 – February 2024 · Pune, India

- Built **3–4 Django modules from scratch**, designing PostgreSQL schemas, REST APIs, serializers, views and business logic.
- Cut SQL execution time from **~15–25 s to ~1–2 s** by optimizing Django ORM and raw SQL, joins, subqueries and indexes.
- Reduced database queries per request from **500+ to ~5–10** by eliminating N+1 queries with `select_related` / `prefetch_related` and fetching only required fields.
- Built asynchronous and scheduled **Celery workflows** for email notifications and reminders, debugging worker execution and failed tasks.
- Modernized components in a **~10-year-old Django codebase**, replacing inefficient implementations to improve maintainability and performance.
- Shipped end-to-end features with **Vue.js, JavaScript and TypeScript**: components, forms, Vuex/Pinia, routing and Django REST API integration.

02 Technical skills

LANGUAGES	Python, SQL, JavaScript, TypeScript
BACKEND	Django, Django REST Framework, REST APIs, Django ORM, API design, authentication and authorization
DATABASES	PostgreSQL, schema design, query optimization, indexing, transactions, JSONB, pgvector
ASYNC & DISTRIBUTED	Celery, background jobs, asynchronous workflows, distributed task execution, multi-tenant systems, microservices
AI ENGINEERING	LLM APIs, LLM routing, RAG, embeddings, vector search, tool calling, agent workflows, multi-provider integration
FRONTEND	Vue.js, Vuex, Pinia, Tailwind CSS, HTML, CSS
TOOLS	Docker, Docker Compose, Redis, Git, GitHub, Jira, Sentry, MailHog

03 Education

Sikkim Manipal Institute of Technology

2017–2021

Bachelor of Technology in Computer Science and Engineering